Artificial Intelligence and Machine Learning (6CS012)

Part - 1

Student Name: Lam Pasang Lama

Student ID: 2408830

Tutor: Jinu Nyachhyon

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Long Question

Selected Question: Data Scientist at a Digital Payment Platform (eSewa)

Introduction

In the rapidly evolving landscape of digital payments, platforms like eSewa process millions of transactions Daily. Leveraging Artificial Intelligence is no longer optional but a necessity for security and growth. While supervised learning thrives on labeled data, unsupervised learning provides critical insights when specific labels, such as "fraud," are missing or rare. This report examines how these techniques solve real-world financial challenges.

Problem Identification

Two major challenges faced by payment platforms are **Fraud Detection** and **Customer Behavioral Profiling**. Fraudulent transactions are often rare anomalies that deviate from a user's typical spending patterns. Identifying these without historical labels is a significant hurdle. Additionally, the platform needs to segment its massive user base—differentiating between utility bill payers and frequent merchant shoppers—to tailor marketing and community donation features like "Helping Hand."

Proposed Approaches

To address fraud detection, I propose using **Auto encoders**. These neural networks are trained to compress and then reconstruct "normal" transaction data. A fraudulent transaction, being an outlier, will result in a high reconstruction error, flagging it for review. For behavioral profiling, I recommend **Clustering** via the **DBSCAN** algorithm. DBSCAN is superior to traditional methods as it can discover clusters of any shape and effectively isolate noisy data, which is common in complex financial datasets.

Implementation and Challenges

These models can be integrated into real-time risk scoring engines. If a transaction yields a high anomaly score, the system can trigger an immediate secondary verification or block the payment. However, a primary challenge is the risk of "False Positives," where legitimate but unusual behavior (e.g., a large one-time donation) might be incorrectly flagged, causing friction in the user experience.

Short Questions

Selected Question: Overfitting and under fitting

Overfitting occurs when a machine learning model learns the training data too well, including its noise and random fluctuations. This results in excellent training performance but poor generalization to new, unseen data (high variance). In contrast, **under fitting** occurs when a model is too simple to capture the underlying patterns in the data, leading to poor performance on both training and test sets (high bias).

The impact on generalization is significant: an over fitted model "memorizes" the past, while an under fitted model fails to learn it. A successful AI implementation requires finding the "sweet spot" where the model generalizes the underlying logic of the data without being distracted by noise.

Selected Question: CNN vs RNN Architecture

Convolutional Neural Networks (CNNs) are specialized for grid-like data, such as images. They use spatial hierarchies and kernels to detect features like edges and textures. **Recurrent Neural Networks (RNNs)** are designed for sequential data, such as natural language or time series, where the current output depends on previous time steps.

Key challenges in deep learning include **Vanishing Gradients**, which can be mitigated using **Batch Normalization** or ReLU activation functions. Furthermore, **Overfitting** in these architectures is commonly managed using **Dropout** layers, which randomly disable neurons during training to ensure the network learns more robust, non-redundant features.

Conclusion

In conclusion, the deployment of machine learning in digital payment systems requires a careful balance between different architectures and learning paradigms. Unsupervised learning through Autoencoders and DBSCAN offers a sophisticated way to handle unlabeled data for security and user segmentation. By effectively managing technical challenges like overfitting and gradient stability, platforms can provide a safer and more personalized experience for their communities.